

THEJUS MAHAJAN PH.D.

Computational Scientist | Machine Learning · Interpretability · Clinical Data

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*30.09.1992 (Talikulam, India)

Married

Nationality: Indian

Arbeitsberechtigung für Deutschland vorhanden · Work authorisation: yes

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Google Scholar



PROFILE

- Computational scientist (PhD, Post-doc) with ten years' experience building and validating computational models — in physics, marine ecosystem modelling, bioinformatics and clinical data.
- Currently conducting independent research in **neural network interpretability** (PyTorch, ONNX, transformer models), alongside applied work in clinical data engineering and biostatistics.
- Seeking roles in machine learning, research software engineering or scientific data science. Available immediately; open to relocation within Germany.

PROFESSIONAL EXPERIENCE

Independent Research — Neural Network Interpretability

Self-directed, open source

07/2026 - present

Hamburg, Germany

Mechanistic interpretability of transformer neural networks

- Built an interpretability toolchain for a 15-layer transformer network (Leela Chess Zero): captures the model's internal attention during inference via **PyTorch** forward hooks on **ONNX**-converted weights.
- Diagnosed and corrected two systematic errors in the analysis pipeline that produced plausible but invalid results — neither raised an exception nor failed a test. Published the analysis, including a public correction of my own earlier findings.
- Implemented batched GPU/CPU inference and asynchronous orchestration of the reference engine, making whole-game analysis feasible.
- Full-stack research application (Python/FastAPI, React/TypeScript) with an automated test suite.
- Write-up: *The Mirror in the Machine* (thejusmahajan.github.io)

Clinical Data Engineering — Praxisphase (Weiterbildung ABI-2025-3)

HealthTwiSt GmbH

02/2026 - 04/2026

Berlin, Germany

Production data engineering for a national medical registry (Dr. A. Busjahn)

- Refactored a production R/tidyverse pipeline for the DeGIR registry (143,000+ patient records, ~300 clinics), reducing the codebase by a quarter with byte-identical output verified at every step.
- Replaced 257 hard-coded rules with a configuration-driven design, moving business logic out of code into reviewable data files.
- Packaged reusable components as a documented R library with automated checks; built an interactive dashboard on a de-identified, perturbed extract rather than patient data.
- Found two pre-existing bugs during code review; documented both and preserved existing behaviour rather than silently changing results.

Continuing Education — Bioinformatics and Biostatistics

CQ Beratung + Bildung GmbH

08/2025 - 02/2026

Berlin, Germany

Applied bioinformatics and biostatistics

- Programming (Python, R, BASH), bioinformatics databases and sequence analysis, NGS pipelines (Nextflow).
- Biostatistics in R: ANOVA, PCA, HCA, multivariate methods.
- Capstone: reproducible Nextflow (DSL2) pipeline for Hepatitis Delta Virus sequence analysis (GitHub).

Guest Scientist

KEY STRENGTHS



Machine Learning & Interpretability

Attention extraction from transformer networks with PyTorch and ONNX. Rigorous validation of model-analysis pipelines: two silent correctness errors found, fixed and published.



Clinical Data Pipeline Engineering

Production R pipeline for a national quality registry (143K records, ~300 clinics), refactored with byte-level verification. R package and Shiny dashboard development.



Scientific Computing & Reproducible Research

High-performance computing (JAX, Linux compute clusters, Fortran to Python migration), reproducible pipelines (Nextflow, Conda), large scientific datasets (NetCDF, HDF5).

TECHNICAL SKILLS

Programming

Python (PyTorch, ONNX, JAX, NumPy, Pandas, FastAPI), R (tidyverse, Shiny, tidymodels), SQL / MySQL, TypeScript, Fortran, BASH

Machine Learning

Transformers, attention/activation capture (forward hooks), mechanistic interpretability, ONNX→PyTorch conversion, batched GPU inference, policy/value head analysis, tidy-models

Clinical Data & Biostatistics

Medical registry pipelines, GDPR/DSGVO compliance, multivariate analysis, PCA, ANOVA, survival analysis concepts, clinical endpoints

Data Engineering

Pipeline refactoring, config-driven ETL, `identical()` verification, R Shiny dashboards, large datasets (HDF5, NetCDF)

Bioinformatics

NGS analysis, Nextflow (DSL2), sequence analysis

Tools

Git/GitHub, Linux/Unix, Linux compute clusters, Conda, LaTeX, Docker concepts

LANGUAGES

English



Fluent (C1)

Helmholtz-Zentrum Hereon, Ecosystem Modelling

📅 05/2025 - 10/2025 📍 Geesthacht, Germany

Marine ecosystem modelling; ran parallel to the continuing education programme from 08/2025. Finalised the computational modelling framework and prepared the simulation data for peer-reviewed publication.

Job Search & German Language Course

📅 02/2025 - 04/2025 📍 Hamburg, Germany

Active job search and German language acquisition (Goethe-Zertifikat B1).

Post-doctoral Scientist

Universität Hamburg, Institute for Marine Ecosystem and Fishery Science

📅 08/2021 - 01/2025 📍 Hamburg, Germany

Marine ecosystem modelling

- Parental leave from 01/2024 to 12/2024.
- Developed the "Cyanobacteria Life Cycle" (CLC) model in two formulations: 0-D agent-based, and 1-D concentration-based (advection–diffusion) within **GOTM-FABM**, for climate-warming projections.
- High-performance computing (HPC): Translation of a legacy Fortran model into a high-performance Python engine (**Google JAX**, GPU parallelisation). Processing of large NetCDF datasets on Linux compute clusters.

Career Transition

📅 04/2021 - 07/2021 📍 Hamburg, Germany

Relocation from India to Germany and active job search.

Physics Tutor & Chess Coach — Freelance / Tutorwaves Solutions Inc., Kerala, India · 10/2018 - 03/2021

Doctoral Researcher in Astrochemistry

Université Paris-Saclay

📅 10/2015 - 09/2018 📍 Orsay, France

Atomic collision experiments, data analysis and modelling on the AGAT setup at the Orsay Tandem accelerator.

- Analysed multi-detector coincidence data from three accelerator campaigns in the DP2 framework, working from NDB-formatted detector output.
- Diagnosed **incomplete-record rates of 15–50%**, traced them to detector positioning, corrected them, and validated the correction by repeating the campaign with a different detector configuration.
- Mass calibration against a tri-alpha source (Pu-239, Am-241, Cm-244; linear fit, $R^2 = 0.99$); timing-based noise rejection; background subtraction by paired control runs; uncertainties quantified per campaign.
- Adapted existing one- and two-electron **Classical Trajectory Monte Carlo** codes to the collision system and ran 10^6 -trajectory simulations with a documented convergence study, at Universidad Autónoma de Madrid and CELIA, Bordeaux.
- Contributed measured dissociation branching ratios and cross sections to the international **KIDA** astrochemical database, for reuse in other groups' chemical models.

German

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B1 (Goethe-Zertifikat), B2 in preparation

French

● ● ● ● ●

Basic knowledge

Malayalam

● ● ● ● ●

Native

Tamil

● ● ● ● ●

Fluent

Hindi

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Basic knowledge

FURTHER TRAINING

Deep Learning with PyTorch

IBM | Coursera — completed 08/2026

Credential DDDI9T0KHUJ4

CNNs, transfer learning, GPU/CUDA training patterns

Deployable Data Analysis & AI Pipelines with HPC

Supercomputing-Akademie, HLRS

Universität Stuttgart | 09–10/2026 (enrolled)

High Performance Computing (HPC)

Jülich Supercomputing Centre (JSC),

Forschungszentrum Jülich

PUBLICATIONS

Four peer-reviewed journal articles and two refereed conference proceedings in atomic and molecular physics / astrochemistry (2018–2020), including *Astronomy & Astrophysics* and *Journal of Physics B*.

🎓 Google Scholar Profile

EDUCATION

Ph.D. in Astrochemistry

Université Paris-Saclay, Institute of Molecular Sciences (ISMO)

📅 10/2015 - 09/2018 📍 Orsay, France

Supervisor: Dr. Karine Béroff. Thesis: *Excitation and fragmentation of C_nN^+ ($n=1-3$) in collisions with He atoms at intermediate velocity.*

Research Project – Theoretical Atomic Physics

IIT Mandi (under Dr. Hari Varma)

📅 02/2015 - 08/2015 📍 Mandi, India

Theoretical atomic physics; partly remote.

M.Sc. in Physics

National Institute of Technology Calicut, India

07/2012 - 12/2014

Calicut, India

First Class with Distinction (CGPA 8.71/10).

Hamburg, 16 August 2026

B.Sc. in Physics

University of Calicut, India

06/2009 - 04/2012

Thrissur, India

Grade: B+ (CGPA 3.49/4.0).



Dr. Thejus Mahajan